Quantum machine learning algorithms

[{'objective': 'Introduce and establish the growing synergy between quantum machine learning (QML) and photonic quantum computing (PQC) by highlighting the benefits and potential applications of PQC.', 'dataset': 'Not found', 'model': 'Not found', 'accuracy': 'Not found', 'limitations': 'Not found', 'paper': 'papers\\MerLin_A_Discovery_Engine_for_Photonic_and_Hybrid_Quantum_Machine_Learning.pdf'}, {'objective': 'To investigate the practical benefits of quantum machine learning for near-term QML, by systematically exploring different quantum model designs across various tasks.', 'dataset': 'Not Found', 'model': 'Not Found', 'accuracy': 'Not Found', 'limitations': 'Not Found', 'paper': 'papers\\MerLin_A_Discovery_Engine_for_Photonic_and_Hybrid_Quantum_Machine_Learning.pdf'}, {'objective': 'Design and implement MerLIN – an open-source framework for systematic and empirical exploration of photonic and hybrid quantum machine learning models and hardware constraints.', 'dataset': 'Reproduces eighteen state-of-the-art photonic and hybrid QML works.', 'model': 'MerLIN includes optimized strong simulation of linear-optical circuits.', 'accuracy': 'Designed around systematic benchmarking and reproducibility.', 'limitations': 'Not Found', 'paper': 'papers\\MerLin_A_Discovery_Engine_for_Photonic_and_Hybrid_Quantum_Machine_Learning.pdf'}, {'objective': "To outline the research's aim to bridge the gap between quantum Machine Learning (QML) and photonic hardware for near-term hybrid systems.", 'dataset': 'The paper primarily focuses on establishing frameworks that facilitate consistent evaluations across platforms for QML models, examining how embedding trainable quantum circuits into standard AI pipelines enables hybrid workflows.', 'model': 'The paper emphasizes the growing landscape of quantum software frameworks (Qiskit, Cirq) and its importance for near-term quantum-inspired AI explorations.', 'accuracy': 'Scalability and benchmark-driven experimentation are highlighted as keys for progress within the QML field.', 'limitations': "The paper's objective is to offer a description and overview of the landscape.", 'paper': 'papers\\MerLin_A_Discovery_Engine_for_Photonic_and_Hybrid_Quantum_Machine_Learning.pdf'}, {'objective': 'Develop a framework for integrating photonic quantum models into machine learning ecosystems and enabling distributed benchmarking.', 'dataset': 'Not Found', 'model': 'Not Found', 'accuracy': 'Not Found', 'limitations': 'Not Found', 'paper': 'papers\\MerLin_A_Discovery_Engine_for_Photonic_and_Hybrid_Quantum_Machine_Learning.pdf'}]

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